Lab Manual: Practical 5

Course Code: ETCS406

Course Title: Game Development

Program: B. Tech Computer Science and Engineering

Semester: VII

Practical Title: Build a Unity-Based Game Prototype Integrating Simple AI (e.g., NPC Using Pathfinding with NavMesh)

Duration: 4 Hours

Lab Type: AI Integration & Scripting

Assessment Weightage: Part of CAL 1 (Lab Assignments)

Objective

To implement basic artificial intelligence in a Unity game prototype using NavMesh for NPC pathfinding. Students will learn how to make non-player characters (NPCs) navigate a game environment intelligently.

Pre-requisites

- Familiarity with Unity Editor and GameObjects
- Basic C# scripting knowledge
- Understanding of Unity's Navigation system (NavMesh)
- Exposure to AI concepts like pathfinding and agent behavior

Learning Outcomes

By the end of this lab, students will be able to:

- Set up Unity's NavMesh system for pathfinding
- Create and configure NPCs with AI navigation behavior
- Use C# scripts to control NPC movement and interaction
- Simulate basic AI-driven gameplay scenarios

Tools & Resources

- Unity Editor (2020.3 LTS or later recommended)
- Visual Studio or any C# IDE
- Unity NavMesh Components (available via Package Manager or GitHub)
- Reference: Unity Learn – Pathfinding with NavMesh

Lab Tasks

Task 1: Scene Setup

- Create a new Unity 3D project
- Design a simple environment with obstacles and walkable areas
- Add a player GameObject and an NPC GameObject

Task 2: NavMesh Configuration

- Mark walkable surfaces using **NavMeshSurface**
- Bake the NavMesh to generate navigation data
- Add **NavMeshAgent** component to the NPC

Task 3: AI Scripting

- Write a C# script to make the NPC follow the player or move to a target location
- Use NavMeshAgent.SetDestination() to define movement goals
- Implement basic obstacle avoidance and movement smoothing

Evaluation Criteria

Component	**Marks**
NavMesh Setup & Environment	3
NPC Configuration & Movement	3
AI Scripting & Behavior Logic	2
Code Quality & Debugging	2
Total	**10**

Tips for Success

- Ensure all walkable surfaces are properly marked before baking NavMesh
- Use **Gizmos** to visualize agent paths and destinations
- Test with different obstacle layouts to observe AI behavior
- Keep scripts modular and well-commented for clarity

Suggested Reading & Resources

- *Introduction to Game Development* by Steve Rabin – Chapters on AI and pathfinding
- Unity Learn: NavMesh and AI Navigation
- Brackeys YouTube: Unity NavMesh Tutorial

Submission Guidelines:

- Submit your report in PDF format via the Google Classroom.
- Include screenshots or gameplay references where relevant.